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Effects of roots systems on hydrological connectivity below the soil surface in the Yellow River Delta wetland

Yinghu Zhang[1,2,3] | Jiang Jiang[1,2,3] | Jinchi Zhang[1,2,3] | Zhenming Zhang[4] | Mingxiang Zhang[4]

[1]Co-Innovation Center for Sustainable Forestry in Southern China, Nanjing Forestry University, Nanjing, China

[2]Jiangsu Provincial Key Laboratory of Soil Erosion and Ecological Restoration, Nanjing Forestry University, Nanjing, China

[3]Priority Academic Program Development of Jiangsu High Education Institutions (PAPD), Nanjing Forestry University, Nanjing, China

[4]School of Ecology and Nature Conservation, Beijing Forestry University, Beijing, China

Correspondence
Mingxiang Zhang and Zhenming Zhang, School of Ecology and Nature Conservation, Beijing Forestry University, Qinghua East Road 35, Haidian District, Beijing 100083, China.
Email: pku2015hold@163.com; zhenmingzhang@bjfu.edu.cn

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Abstract

In this study, a new method based on field dye-tracing experiments was used to assess hydrological connectivity below the soil surface. In this method, *Phragmites australis* in the Yellow River Delta wetland was considered to characterize the interaction between roots systems and hydrological connectivity. The results showed that index of hydrological connectivity (*IHC*) declined non-linearly with depth below the soil surface. Roots systems of *P. australis* were mostly distributed in soils of 0–30 cm. Roots systems parameters (root length, root width, root surface area, root projected area, root volume and root biomass) were positively correlated with *IHC*. Coarse roots systems (3 < D < 5 mm) made the highest contribution to the changes of *IHC*. This study provided suggestions for wetland management in the study area: (1) Reeds should not be reaped every year to promote the litters decomposition for roots nutrition availability; and (2) freshwater inputs should be strengthened in the initial/middle growth stage of reeds to improve roots architecture. Knowledge about the interaction between roots systems and hydrological connectivity below the soil surface is crucial for large-scale wetland protection as wetland managers regulate the control and distribution of vegetation.

KEYWORDS

below the soil surface, hydrological connectivity, roots systems, wetland protection, Yellow River Delta wetland

1 | INTRODUCTION

Coastal wetlands provide multiple ecosystem services, such as carbon storage, species habitat, groundwater recharge and water quantity and quality (Cui et al., 2016; Liu et al., 2021). However, they are under tremendous pressure from anthropogenic activities (Schuyt, 2005) and climate change (Zhang, Li, et al., 2019). These pressures influence the functions of coastal wetlands through daily, seasonal and annual dynamic changes to water levels (Galatowitsch, 2016) and ultimately cause wetland degradation (Wang et al., 2020). Currently, many countries of the world are exposed to the threat of wetland degradation (Deng, 2020). In China, the area of coastal wetland declined by 51% in the last half century (Liu et al., 2019). Thus, coastal wetland protection has become important support direction for governments (Liu et al., 2021). The Yellow River Delta is one of the priority areas and the most active deltas in China (Liu et al., 2019). It has been on the list of Ramsar Wetlands of International Importance since 2013 (Zhao et al., 2019). It plays an important role in water purification, biodiversity maintenance and bird migration (Chen et al., 2016). In addition to its ecological functions, it also has economic development potential (Bai et al., 2019). During the last several decades, the delta has experienced changes attributed to the agricultural exploration, urban residential area expansion and petroleum refining. Those environmental changes cause the deterioration of the structure and function of wetland ecosystems (Lu et al., 2021), especially wetland degradation and disappearance (Deng, 2020; Meng et al., 2019). Wetland degradation

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in this area enhances the complexity and spatial heterogeneity of isolated wetlands with the surrounding water bodies, which results in negative hydrological responses (Zhu et al., 2020). Hydrological responses as the most fundamental element of wetland ecosystems will affect its stability and sustainability (Meng et al., 2019).

Recently, the term 'hydrological connectivity' provides a new perspective to indicate the changes of hydrological responses and to define its correlations with ecological functions (Liu et al., 2019; Zhang et al., 2021). Hydrological connectivity defines the water-mediated transfer of mass, energy and organisms from one part of the landscape to another at larger scales (Bracken & Croke, 2007; Xie et al., 2021). The dynamic changes of hydrological connectivity caused by volatile hydrological conditions have vital importance in wetland protection (Liu et al., 2020). There have been surprisingly multiple studies reporting hydrological connectivity above the soil surface for managers or governments to restore wetlands or create new wetlands (Chen et al., 2016; Jiang et al., 2021; Zhao et al., 2019). For example, high hydrological connectivity above the soil surface will disrupt the survival of the autochthonous producers in coastal salt marshes and damage wetland ecosystem services (Yin et al., 2020), whereas Wang et al. (2021) found that hydrological connectivity above the soil surface could improve soil physical and chemical properties, such as reducing soil compactness and soil salinity and increasing soil water content. However, our understanding of hydrological connectivity below the soil surface remains an extreme gap in our recent works. Hydrological connectivity below the soil surface is gradually being recognized as an important aspect of connectivity (Dai et al., 2020). It is well established that hydrological connectivity below the soil surface may strongly affect hydrological responses, such as preferential water flow directly connected to the surface or subsurface (Borges et al., 2019; Jarvis et al., 2017). Those flow paths function well vertically and horizontally below the soil surface (Zhang et al., 2018) and affect water, solutes and energy transport in the rhizosphere (Borges et al., 2019; Jarvis et al., 2017). It is reasonable for evaluating hydrological connectivity below the soil surface since the assessments are very crucial to larger scale wetland management (Koestel et al., 2013; Vogel & Roth, 2003). It must not be forgotten that hydrological connectivity in the rhizosphere need to be scaled up if they might be generalized at larger scales (Koestel et al., 2013). The active reaction zone, that is, rhizosphere, is induced by the interaction of roots systems and hydrological responses (Stottmeister et al., 2003). Preferential roots channels during its decomposition generate a vertical and horizontal connected transport path network (Zhang et al., 2017, 2018). However, the interaction between the architecture of roots systems and hydrological responses remains a gap. Understanding about the interaction is crucial for large-scale wetland protection and management.

In this study, we used field dye-tracing experiments methods for (1) the assessment of index of hydrological connectivity below the soil surface for *Phragmites australis* wetland; (2) the quantification of the architecture of roots systems; and (3) the effects of roots systems parameters on *IHC*. The understanding of index of hydrological connectivity below the soil surface is the first step for wetland protection and management. The newly developed index of hydrological connectivity in the current work is expected to be the foundation of future work. Based on the results, we proposed an alternative wetland management solution in the Yellow River Delta wetland. Knowledge of the newly assessment of hydrological connectivity below the soil surface in this area can help other regions avoid similar problems and improve the protection of coastal wetland effectively.

2 | MATERIALS AND METHODS

2.1 | Study area

The study area is located in the Yellow River Delta National Natural Reserve (118°32′981″–119°20′450″E, 37°34′768″–38°12′310″N) in north-eastern Shandong Province, China (Figure 1). The Yellow River Delta National Natural Reserve covers an area of 153 km^2. The main landform types of the study area are the land zone, tidal flat zone and subtidal zone. The climate is warm-temperate continental with a mean annual precipitation of 576.7 mm, a mean annual potential evapotranspiration of 1,962 mm, a mean annual temperature of 12.1°C and the frost-free period 196 d. The main soil type in the study area is mainly coastal saline soil, derived from the sediment (Zhang et al., 2016). The three main vegetation types are deciduous broad-leaved forests, marsh vegetation and salt-bearing vegetation. The

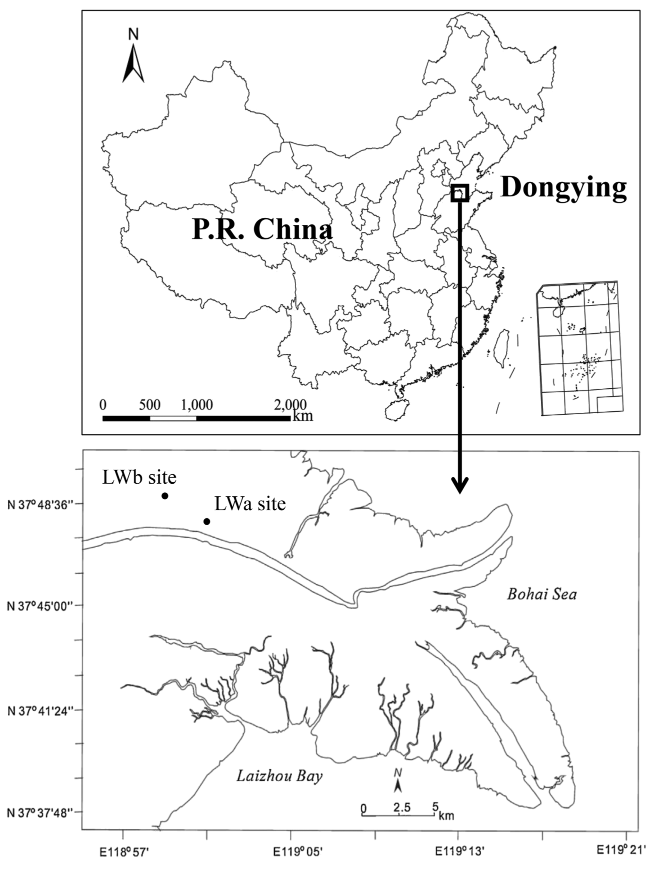

FIGURE 1 Location of the study area

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dominant vegetation is *Robinia pseudoacacia* in the deciduous broad-leaved forest communities, *P. australis* in the marsh vegetation communities and *Tamarix chinensis* and *Suaeda salsa* in salt-bearing vegetation communities.

2.2 | Roots systems acquisition

In wet season from July to August 2017, the sample sites were chosen in *P. australis* wetlands. Two different sites (i.e. large [LWa] and small communities [LWb] of *P. australis*) with a flat topography were located at the Yellow River Delta wetland (Figure 1). In this study, the LWa sites are areas with a higher above-ground dry weight (1000 g), coverage (98%) and height (2.0 m), whereas the LWb sites are typical areas with a lower above-ground dry weight (200 g), coverage (40%) and height (0.6 m). Three experimental plots with similar site conditions were randomly selected as replicates from the 10.0 × 10.0-m quadrat each site, and there are total six plots studied.

At each sample experimental plot, soil cores (7 cm in diameter and 10.4 cm in height, 400 cm^3) with five replicates taken at 10-cm intervals were collected. Soil samples were removed from the soil cores, placed in plastic bags, numbered and refrigerated (4°C) until the separation of roots systems. When necessary, soil samples were placed in dishes with 4–5 mm of water for 2 h so that the roots systems could spread and the soil particles could be easily removed, and then the soil was separated from the roots systems under running water using 5-mm mesh sieves. Roots systems were classified according to the roots systems diameters (D < 1, 1 < D < 3, 3 < D < 5, and D > 5 mm), dried for 48 h in an oven at 70°C and then analysed using RootAnalysis professional software (PM-Tech GmbH, Germany) to determine the architecture of roots systems, that is, root length (RL), width (RW), surface area (RSA), projected area (RPA), volume (RV) and biomass (RB). The values of roots systems parameters are concluded in Table 1.

2.3 | Field methodology

In this study, field dye-tracing experiments (Figure 2) were conducted to provide useful information for the assessment of index of hydrological connectivity below the soil surface. The field methodology is resource effective and has few restrictions in operation as well as low adsorption and high mobility (Guo et al., 2019; Wu et al., 2021). At each experimental plot, a 1.2 × 1.2-m area was designated for field experiments. A Brilliant Blue FCF (C.I. Food Blue 2) dye solution (4 g/L) with a total volume of 50 L was uniformly applied to the area of the experimental plot. Horizontal soil profile intervals of 10 cm based on the maximum dye depth were excavated 1 day after the dye solution application (Figure 2). The horizontal soil profiles in the two sites were prepared for photographing with a digital camera. The stained horizontal profiles were prepared for the processes of enhancing image contrast, image de-noising and image binarization with the software ImageJ 1.8.0 Fiji (Yi et al., 2020). The pixels in each binarized image were counted using MATLAB (Dai et al., 2020). Dye coverage (*DC*) and fractal dimension (*FD*) were determined by ImageJ.

- *DC* is defined as the percentage ratio of the dye-stained areas to the total horizontal soil profile areas.

$$DC = [D/(D+ND)] \cdot 100\%$$

where *D* is the dye-stained area and *ND* is the non-dye-stained area.

- *FD* characterizes the complexity of dye-stained areas in the horizontal soil profiles. The larger the *FD*, the more complex the dye-stained areas.

$$FD = \lim_{\varepsilon \to 0} [\log N(\varepsilon)/\log(1/\varepsilon)]$$

where ε is the average width of each dye-stained area and $N(\varepsilon)$ is the number of fractals covered by this dye-stained area.

2.4 | *IHC* characterization

The *IHC* was derived based on the fraction of the horizontal sectional area that was stained with the dye. The use of *IHC* over *DC* should be

TABLE 1 Roots systems parameters in the two sites

Sites	SD (cm)	RL ($mm/100cm^3$)	RW ($mm/100cm^3$)	RSA ($mm^2/100cm^3$)	RPA ($mm^2/100cm^3$)	RV ($mm^3/100cm^3$)	RB ($g/100cm^3$)	VC (%)
LWa	0–10	2443.99 ± 183.20	112.89 ± 19.07	4017.96 ± 305.71	1268.61 ± 100.34	2695.25 ± 579.60	0.45 ± 0.20	98
	10–20	1620.34 ± 452.44	78.11 ± 17.68	2877.35 ± 636.60	906.68 ± 204.84	1945.89 ± 630.66	0.25 ± 0.07	
	20–30	818.37 ± 781.80	31.51 ± 28.56	1019.54 ± 900.06	323.31 ± 285.27	307.43 ± 287.13	0.27 ± 0.09	
LWb	0–10	1743.73 ± 99.54	87.05 ± 25.81	3763.27 ± 1100.05	1188.02 ± 343.68	3059.05 ± 2146.97	0.47 ± 0.12	40
	10–20	1044.90 ± 522.18	44.41 ± 12.92	2065.31 ± 1132.75	651.98 ± 358.47	1688.06 ± 1048.75	0.25 ± 0.03	
	20–30	760.18 ± 314.52	37.58 ± 15.05	1889.46 ± 1365.76	600.10 ± 434.31	1304.06 ± 1067.70	0.18 ± 0.09	

Abbreviations: RB, root biomass; RL, root length; RPA, root projected area; RSA, root surface area; RV, root volume; RW, root width; SD, soil depth; VC, vegetation cover.

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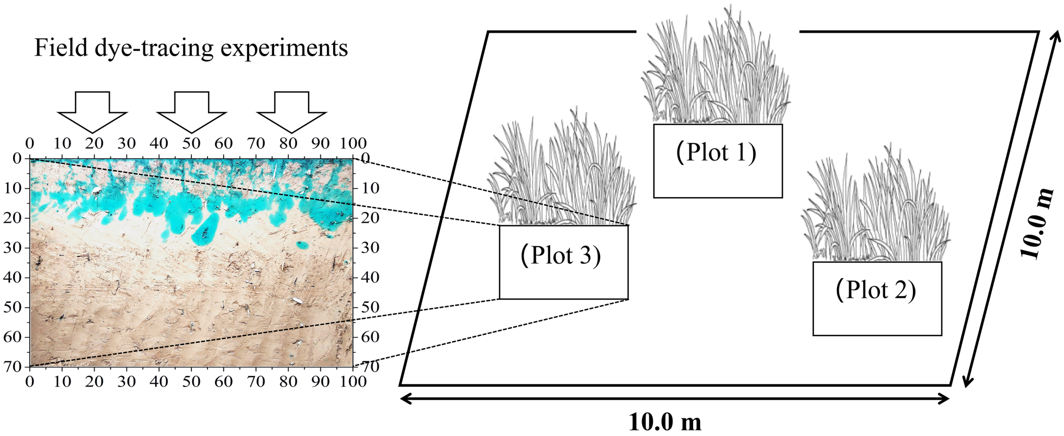
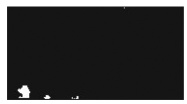
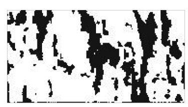
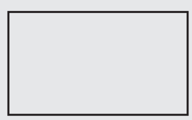


FIGURE 2 Field dye-tracing experiments

TABLE 2 *IHC* evaluation and description

IHC	Interaction between preferential water flow and the surrounding soil matrix	DC	Description
1.00	Homogeneous soil matrix	0.95–1.00	
0.95		0.90–0.95	
0.90		0.85–0.90	
0.85		0.80–0.85	
0.80	Heterogeneous soil matrix	0.75–0.80	
0.75		0.70–0.75	
0.70		0.65–0.70	
0.65		0.60–0.65	
0.60	High	0.55–0.60	
0.55		0.50–0.55	
0.50		0.45–0.50	
0.45		0.40–0.45	
0.40	Mixed (high and low)	0.35–0.40	
0.35		0.30–0.35	
0.30		0.25–0.30	
0.25		0.20–0.25	
0.20	Low	0.15–0.20	
0.15		0.10–0.15	
0.10		0.05–0.10	
0.05		0.00–0.05	
0.00	No	0.00	

justified (Table 2). However, *FD* should also be used when the *DC* across the horizontal soil profiles is the same. Though the dye-stained areas are the same, *IHC* may not be equal due to the staining patch fragmentation. If the *DC* of the horizontal sectional area is the same, more complex staining area patches are confirmed when the *FD* increases. A large number of complex staining patches imply low connectivity of each staining patch. On the contrary, few complex staining patches imply high connections for each staining patch.

To better characterize the degree of hydrological connectivity below the soil surface, we calculated *IHC* based on *DC* and *FD*. Before

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we quantify *IHC*, we should indicate the relationship between *DC* and *FD*. In this study, if the *DC* of each soil layer was 0, we defined *IHC* as 0, and if *DC* was 1, *IHC* was 1 (Table 2). First, we quantified the *DC* of each soil layer and characterized the changes in *DC* with the increase in soil depth. *FD* had a high correlation with *DC* (Table 3 and 4); thus, we constructed a quantifiable relationship between *DC* and *FD*. Second, we indicated the results of *IHC* corresponding to *DC* in Table 2. Finally, a well-defined *IHC* was calculated. *IHC* was in the range [0,1]. The different *IHC* in this study implied the degree of hydrological connectivity. The six levels of hydrological connectivity are listed in Table 2. If *IHC* was low, more time was required to balance the hydraulic pressure between the dye-stained areas and non-dye-stained areas. More dye solution could be transported through preferential water flow paths (Zhang et al., 2018). If *IHC* was high, it required little time to balance the hydraulic pressure between the two zones.

2.5 | Statistical analysis

One-way analysis of variance (ANOVA) was used to compare the differences in roots systems and *IHC* at each soil layer between LWa site and LWb site. The Pearson correlation coefficient was conducted using SPSS 22.0 software package to determine the interaction between roots systems and *IHC*. The results were reported at a 5% significance level.

3 | RESULTS AND DISCUSSIONS

3.1 | *IHC* calculation-based *DC* and *FD* of horizontal sectional profiles

The results indicated that *DC* declined non-linearly with the increase in soil depth in both LWa site and LWb site (Table 3 and 4). *FD* increased at first and then declined with an increase in *DC* (Table 3 and 4). The results showed that *IHC* declined non-linearly with increasing soil depth (Figure 3). With respect to surface soils (0–10 cm), *IHC* was 0.94 ± 0.02 in LWa site and 0.60 ± 0.03 in LWb site. For soil depth of 10–20 cm, *IHC* was 0.63 ± 0.09 in LWa site and 0.30 ± 0.09 in LWb site. With respect to the deeper soils (20–30 cm), *IHC* was 0.34 ± 0.06 in LWa site and 0.18 ± 0.04 in LWb site. For the rhizosphere (0–30 cm from the soil surface), *IHC* was higher in LWa site (0.63 ± 0.30) than in LWb site (0.36 ± 0.22). A higher *IHC* in LWa site suggests high efficiency in balancing hydraulic pressure between the dye-stained areas and non-dye-stained areas, which is consistent with the results of Dai et al. (2020) and Wu et al. (2021). In LWa site, more litter and thick humus on the soil surface reduced splash erosion and soil crusting. A higher abundance of roots systems (Table 1), pore formation at the interface between the roots systems and the soil and preferential channels formed by roots systems decomposition improved hydrological connectivity. However, LWb site had less litter and humus on the soil surface which did not prevent the soil structure from being destroyed. The seriously compacted soil surface in LWb

TABLE 3 Changes of *DC*, *FD* and *IHC* with soil depth in LWa site

Plots	Soil depth (cm)	DC	FD	Defined *IHC*	Modelling *IHC*
LWa1	0	0.965	5.36	1.00	0.996
	5	0.925	98.37	0.95	0.954
	10	0.807	209.87	0.85	0.832
	20	0.339	212.75	0.35	0.352
	30	0.056	119.38	0.10	0.084
	40	0.061	57.99	0.10	0.089
	50	0.046	19.25	0.05	0.075
LWa2	0	0.997	1.00	1.00	1.029
	5	0.644	332.54	0.65	0.663
	10	0.480	316.62	0.50	0.493
	20	0.237	281.33	0.25	0.253
	30	0.090	165.13	0.10	0.115
	40	0.495	69.46	0.50	0.509
	50	0.000	0.00	0.00	0.033
LWa3	0	0.970	1.05	1.00	1.001
	5	0.828	189.4	0.85	0.854
	10	0.578	332.95	0.60	0.594
	20	0.491	97.48	0.50	0.504
	30	0.029	19.77	0.05	0.060
	40	0.001	2.22	0.05	0.034
	50	0.000	0.00	0.00	0.033

TABLE 4 Changes of *DC*, *FD* and *IHC* with soil depth in LWb site

Plots	Soil depth (cm)	DC	FD	Defined *IHC*	Modelling *IHC*
LWb1	0	0.998	1.00	1.00	1.007
	1	0.756	28.09	0.80	0.768
	5	0.225	175.27	0.25	0.248
	10	0.370	137.81	0.40	0.392
	20	0.207	35.97	0.25	0.229
	30	0.010	2.09	0.05	0.041
	40	0.004	5.50	0.05	0.035
	50	0.041	3.34	0.05	0.070
LWb2	0	0.981	5.41	1.00	0.990
	1	0.845	26.86	0.85	0.856
	5	0.337	128.92	0.35	0.360
	10	0.337	62.11	0.35	0.360
	20	0.377	33.54	0.40	0.399
	30	0.124	27.93	0.15	0.147
	40	0.022	5.42	0.05	0.052
	50	0.000	0.00	0.00	0.031
LWb3	0	0.994	1.01	1.00	1.003
	1	0.890	19.27	0.90	0.900
	5	0.172	239.66	0.20	0.194
	10	0.172	132.98	0.20	0.194
	20	0.199	24.81	0.20	0.221
	30	0.025	9.63	0.05	0.055
	40	0.000	0.00	0.00	0.031
	50	0.000	0.00	0.00	0.031

site led to greater surface runoff instead of higher infiltration into the soil. In particular, the soil water content in LWb site was lower than that in LWa. The lower soil water content in LWb site may have caused an obvious soil water repellency (Benito Rueda et al., 2016; Rodríguez-Alleres et al., 2012). Soil water repellency contributed more to the high spatial heterogeneity of water flow paths, which showed complicated dyed patches and a variable *IHC* at each soil depth in LWb site (Alanís et al., 2017).

The results indicated that the same *DC* with homogeneous and isolated dye-stained patches would lead to variable *FD*, which results in the changes of *IHC* (Figure 4). Geographically isolated patches in wetland soils may greatly shorten the retention time of pollutants in wastewater (Zhu et al., 2020). *P. australis* wetlands in LWb site with complex isolated patches enhance spatial heterogeneity of hydrological connectivity (Zhu et al., 2020). Pollutants in sewage would transport through larger preferential water flow paths, which reduce the efficiency of wetland sewage treatment (Parsons et al., 2006). Those larger paths attribute from soil fissure development and soil water deficiency in LWb site. Soil fissures can break down hydrological responses and even have negative influences on hydrological connections (Kim & Mohanty, 2017). Re-establishing hydrological connectivity through the construction of a seepage layer from a soil depth of 0–10 cm in LWb site would decrease surface runoff generation (Wu et al., 2020). Shallow ploughing treatment in LWb site would increase topsoil hydrological connectivity, which decreases the connectivity between surface water and groundwater and also decreases the risk of groundwater (Wu et al., 2020). Furthermore, adding litter to the soil surface of LWb site would also improve topsoil properties (Zhang, Xu, et al., 2019). In both LWa site and LWb site, appropriate freshwater should be conducted to the wetlands to alleviate water deficiency. Freshwater input would create a continuous horizontal subsurface flow wetland (Cui et al., 2009). The increase of hydrological connectivity below the soil surface may contribute to the horizontal continuity which prevents wetland clogging, thereby reducing preferential water flow paths and increasing sewage purification efficiency (Muñoz et al., 2006).

3.2 | Determination of roots systems and their relationship with *IHC*

The values of roots systems parameters for each soil depth in LWa site and LWb site are indicated in Table 1. Root length at each soil depth was higher in LWa site than in LWb site ($P > 0.05$), and root length decreased with increasing soil depth. The root length from a soil depth of 0–10 cm was the highest in LWa site (2443.99 ± 183.20 mm/100 cm^3) and LWb site (1743.73 ± 99.54 mm/100 cm^3). In LWb site, root length from soil depth of

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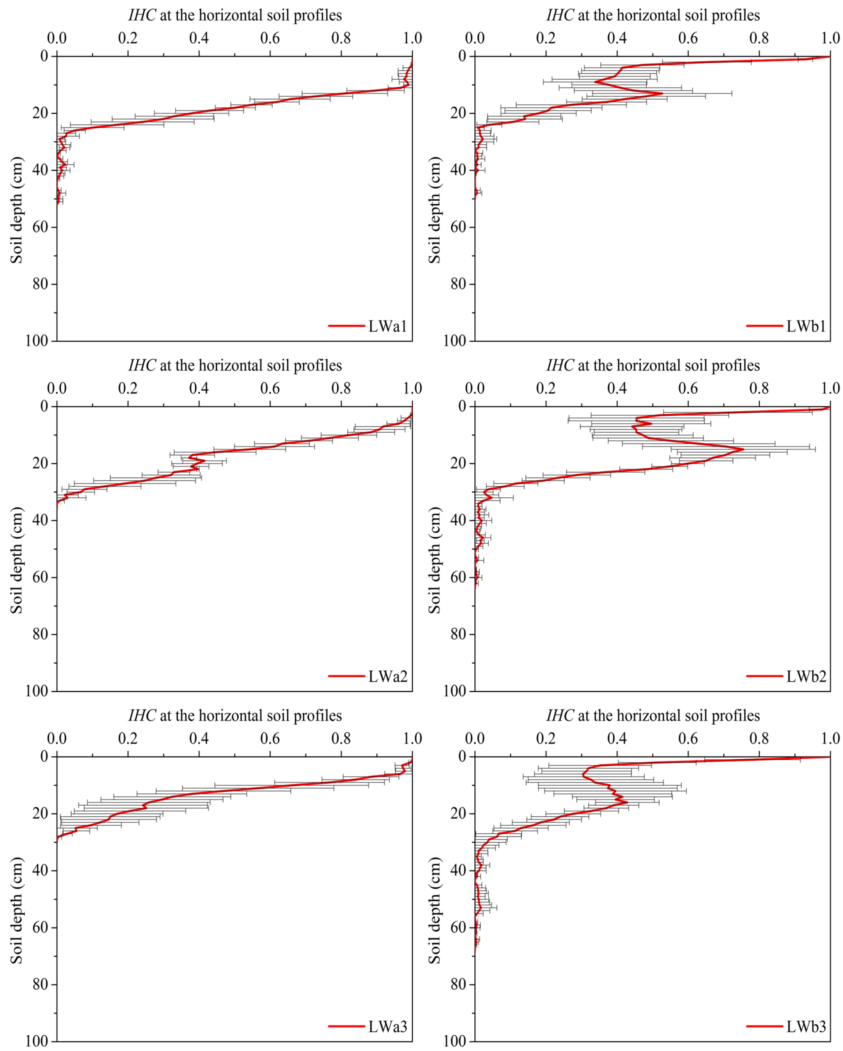

FIGURE 3 Changes in *IHC* with soil depth in the two sites

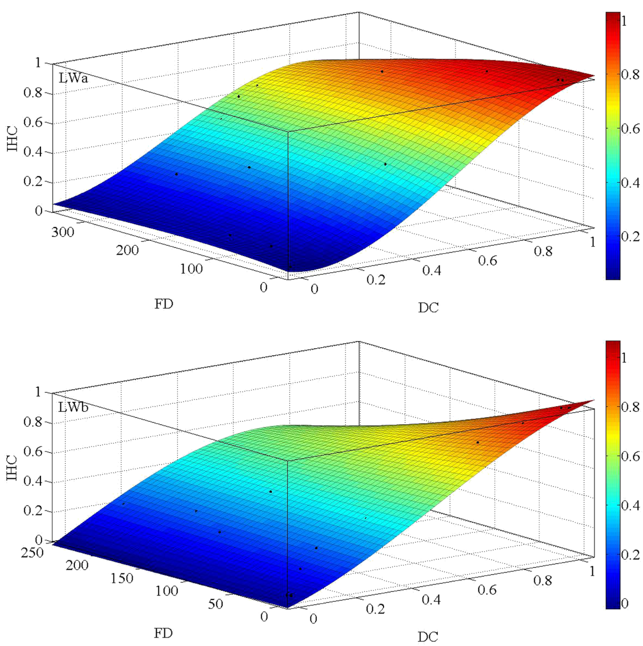

FIGURE 4 Relationship among *DC*, *FD*, and *IHC* in the two sites

0–10 cm was significantly different from those from soil depths of 10–20 cm and 20–30 cm ($P < 0.05$). However, the root length at each soil depth was not significantly different in LWa site ($P > 0.05$). Root width at each soil depth was higher in LWa site than in LWb site ($P > 0.05$), and root width decreased with increasing soil depth. In both LWa site and LWb site, the root width at each soil depth was not significantly different ($P > 0.05$). The root width from a soil depth of 0–10 cm was the highest in LWa site (112.89 ± 19.07 mm/100 cm^3) and LWb site (87.05 ± 25.81 mm/100 cm^3). The root surface area at each soil depth was higher in LWa site than in LWb site ($P > 0.05$). The root surface area from a soil depth of 0–10 cm was the highest in LWa site (4017.96 ± 305.71 mm^2/100 cm^3) and LWb site (3763.27 ± 1100.05 mm^2/100 cm^3). The root surface area at each soil depth was not significantly different in both LWa site and LWb site ($P > 0.05$). The pattern of the root projected area was similar to that of the root surface area. Root volume was the highest from a soil depth of 0–10 cm in LWa (2695.25 ± 579.60 mm^3/100 cm^3) and LWb site (3059.05 ± 2146.97 mm^3/100 cm^3). In both LWa site and LWb site, root volume from a soil depth of 0–10 cm was not significantly different from those from soil depths of 10–20 cm and 20–30 cm ($P > 0.05$). Root biomass was the

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highest from a soil depth of 0–10 cm in LWa site (0.45 ± 0.20 g/100 cm^3) and LWb site (0.47 ± 0.12 g/100 cm^3). Root biomass at each soil depth was not significantly different between LWa site and LWb site ($P > 0.05$).

The results indicated that root length, width, surface area, projected area, volume and biomass were positively correlated with *IHC* (Figure 5), which is consistent with the results of Bodner et al. (2014), Guo et al. (2014) and Hu et al. (2019). In general, roots systems are mostly observed in surface soils instead of deeper soils (Colombi & Keller, 2019). Roots systems in surface soils formed a complex and connected root network during roots decomposition. After roots decomposition, these roots create small pores and biopreferential roots channels with high hydrological connectivity (Bodner et al., 2014). These roots channels can connect with the existing surrounding pores to improve hydrological connections (Guo et al., 2014). Root growth into pre-existing pores leads to the division of large pores into small pores. Our results suggested that root width was more important than other roots systems parameters in affecting the index of hydrological connectivity (Table 5). Nonetheless, other studies have suggested great potential for root surface area to affect decontamination efficiency of wetlands (Kong et al., 2009; Kyambadde et al., 2004). A previous study indicated that wetland performance was strongly correlated with root biomass (Breen & Chick, 1995).

In fact, our results indicated that coarse roots systems (3 < D < 5 mm) in both LWa site and LWb site had a larger influence on the index of hydrological connectivity (Table 5), which is consistent with the results of Cheng et al. (2009). Coarse roots systems had a high degree of lignification. The surfaces of the coarse roots systems might have been sufficiently rough, which increased the resistance to dye solution transport and friction with the surrounding coarse roots systems. Therefore, the phenomenon of the 'water accumulation' in the dye solution due to the formation of macropores was caused by the contact between the rough coarse root surface and the surrounding soils. This phenomenon prompted the simultaneous openings of the horizontally and vertically connected macropore networks surrounding the coarse roots systems. The paths of the dye solutions in the networks were connected. Meanwhile, we concluded that the structure of the surrounding soils was usually disturbed during coarse root growth. Thus, unstable, discontinuous and uneven environmental media were produced because of the contact between coarse roots systems and the surrounding soils. The rough convexity of the coarse roots systems surface and the tortuosity of the coarse roots systems branching increased the disturbance to the surrounding soils during the stretching and thickening processes of the coarse roots systems. It is possible that coarse roots systems might generate large air gaps between the roots and the soils in the case of drought to enhance

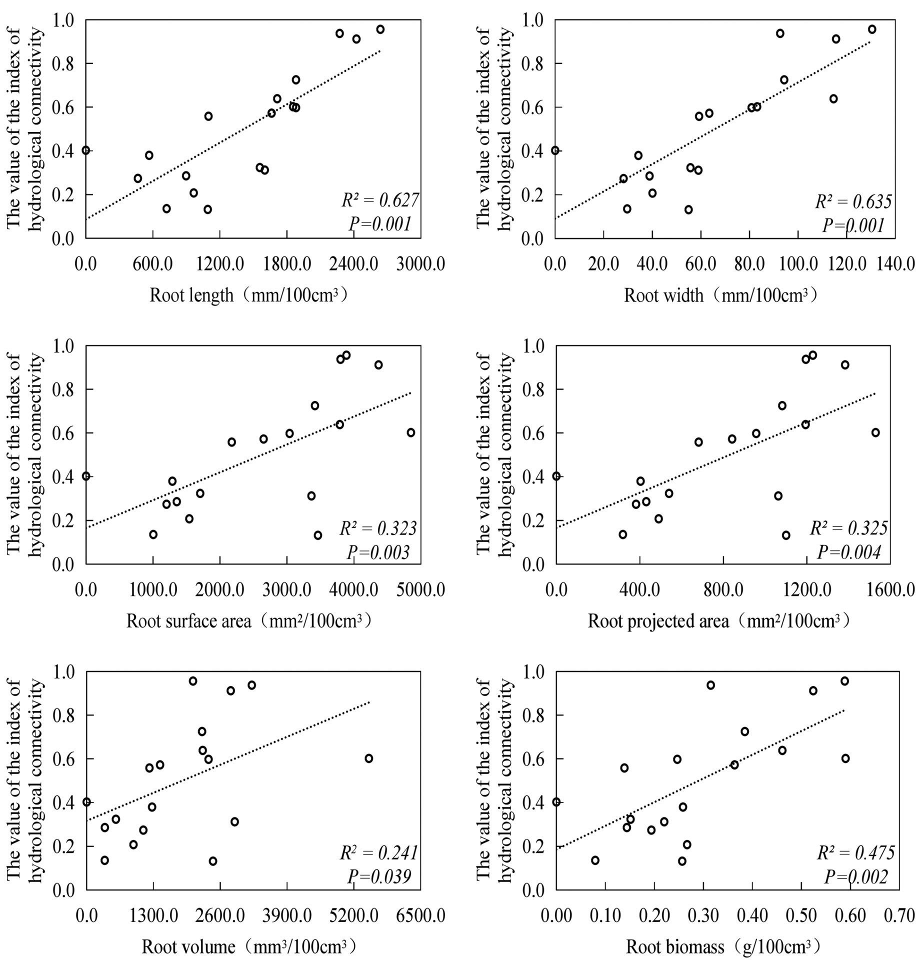

FIGURE 5 The relation between roots systems parameters and *IHC*

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TABLE 5 Pearson correlation coefficients between roots systems parameters and *IHC*

Sites	Roots systems parameters	Roots systems diameter (mm)			
		D < 1	1 < D < 3	3 < D < 5	D > 5
LWa	RL	0.79*($P = 0.01$)	0.71*($P = 0.03$)	0.82**($P = 0.01$)	0.67*($P = 0.05$)
	RW	0.80**($P = 0.01$)	0.79*($P = 0.01$)	0.80**($P = 0.01$)	0.73*($P = 0.03$)
	RSA	0.70*($P = 0.04$)	0.71*($P = 0.03$)	0.83**($P = 0.01$)	0.72*($P = 0.03$)
	RPA	0.70*($P = 0.04$)	0.71*($P = 0.03$)	0.83**($P = 0.01$)	0.72*($P = 0.03$)
	RV	0.63($P = 0.07$)	0.72*($P = 0.03$)	0.84**($P = 0.01$)	0.74*($P = 0.02$)
LWb	RL	0.73*($P = 0.03$)	0.05($P = 0.90$)	0.76*($P = 0.02$)	0.15($P = 0.70$)
	RW	0.81**($P = 0.01$)	0.09($P = 0.81$)	0.71*($P = 0.03$)	0.42($P = 0.26$)
	RSA	0.63($P = 0.07$)	0.05($P = 0.90$)	0.76*($P = 0.02$)	0.32($P = 0.41$)
	RPA	0.63($P = 0.07$)	0.05($P = 0.89$)	0.76*($P = 0.02$)	0.30($P = 0.43$)
	RV	0.38($P = 0.31$)	0.04($P = 0.92$)	0.75*($P = 0.02$)	0.35($P = 0.35$)

Abbreviations: RL, root length; RPA, root projected area; RSA, root surface area; RV, root volume; RW, root width.

water penetration (Bodner et al., 2014). In brief, coarse roots systems with high stiffness induce strong movement of soil particles, which increases the interaggregate void space (Bodner et al., 2014). In addition, the disruption of macroaggregates due to the shifting of coarse roots systems might also form new pores inside the aggregate. Although roots systems radial expansion occurs during root growth, root-induced compaction can increase unsaturated water flow in the soils near the roots systems. Connectivity between soil aggregates increases due to root-induced compaction (Aravena et al., 2011, 2014). These properties of coarse roots systems increased hydrological connectivity to some extent. In particular, the phenomenon of 'water accumulation' in the coarse roots systems caused more dye solution to concentrate around the coarse roots systems. Roots systems increased hydrological connectivity of the rhizosphere and ultimately improved purification efficiency of wetland treatment. However, Cheng et al. (2009) found that fine root biomass might be more important than total root biomass in wetland pollutants removal efficiency. This is because different wetland vegetation (i.e. fibrous-root plants and rhizomatic-root plants) show different roots systems distribution. Fibrous-root vegetation have significantly greater root biomass (D < 1 mm) and a larger root surface area than the rhizomatic-root vegetation (Cheng et al., 2009). *P. australis* as one kind of rhizomatic-root vegetation which generally contain coarse roots systems (D > 3 mm) is more strongly correlated with decontamination (Cheng et al., 2009). Studies showed that lateral and main roots of wetland vegetation could accumulate the highest pollutants and contaminants than leaves, shoots and rhizomes (Cheng et al., 2002). To better promote the highest contribution of *P. australis* to purification efficiency of wetland treatment, appropriate freshwater inputs should also be conducted to the wetlands to decrease soil salinity (Cui et al., 2009). This is because *P. australis* could increase constantly as soil salinity was reduced (Cui et al., 2009). The effects of increasing *P. australis* on wetland function and structure, especially some fibrous-root vegetation introduction for wetland protection and management, require further investigation.

4 | CONCLUSIONS

Our study highlighted that roots systems had a strong influence on hydrological connectivity below the soil surface in *P. australis* wetland of eastern China. First, index of hydrological connectivity (*IHC*) declined with increasing soil depth. Second, roots systems of *P. australis* were rich in soils of 0–30 cm. Third, coarse roots systems (3 < D < 5 mm) had a larger influence on the changes of *IHC*.

The understanding of hydrological connectivity below the soil surface is the first step for wetland protection and management. Based on the results, we proposed an alternative wetland protection solution in this area: (1) Reeds should not be reaped every year to promote the litters decomposition for roots nutrition availability; and (2) freshwater inputs should be strengthened in the initial/middle growth stage of reeds to improve root biomass. Knowledge of the newly assessment of the interaction between roots systems and hydrological connectivity in this area can help other regions avoid similar problems. Additional work is necessary to address the changes in the architecture of roots systems on spatial and temporal scales.

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CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could appear to influence the work reported in this paper.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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ORCID

Mingxiang Zhang https://orcid.org/0000-0002-5972-915X

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